🧬 ATLAS

Product Requirements Document (Final Production Version)

Version: 2.0 (Full Platform Release)
Owner: Dev Mehta
Category: AI for Science / Bioinformatics / Causal Simulation
System Type: Research + SaaS Platform

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1. Executive Overview

ATLAS is a scalable, intervention-aware causal simulation platform for modeling single-cell dynamics under explicit gene perturbations.

It integrates:

- Multimodal single-cell omics data
- Biologically constrained graph reasoning
- Causal intervention modeling (do-operator inspired)
- Graph neural network–based temporal simulation
- Uncertainty estimation and calibration
- Explainability and attribution
- Active learning for experiment optimization
- Web-based interactive visualization
- Containerized reproducible deployment

ATLAS transforms static single-cell analysis into a simulation-first, decision-driven intelligence system.

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2. Vision Statement

To create a unified causal AI framework that enables researchers to simulate, understand, and optimize biological experiments before executing them in the lab.

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3. Strategic Objectives

1. Enable intervention-aware modeling of cellular systems
2. Provide temporally simulated cell-state trajectories
3. Quantify predictive uncertainty
4. Deliver interpretable gene & pathway explanations
5. Recommend optimal next experiments
6. Provide a modular and extensible architecture
7. Support reproducible research workflows
8. Offer SaaS-ready deployment

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4. Problem Definition

4.1 Current Limitations

- Correlation-based analysis dominates bioinformatics
- No explicit modeling of gene interventions
- Temporal dynamics poorly captured
- Lack of uncertainty awareness
- Black-box predictions
- Inefficient experimental planning
- No unified decision-support platform

4.2 Research Gap

There is no integrated framework combining:

- Causal reasoning
- Graph-based biological modeling
- Temporal rollout simulation
- Uncertainty calibration
- Explainability
- Active experiment recommendation

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5. Full System Scope

Included Capabilities

A. Data Layer

- scRNA-seq ingestion
- ATAC-seq integration
- Multi-omics fusion
- Batch correction
- Gene filtering & normalization
- Graph construction from regulatory databases

B. Representation Layer

- Transcriptomic encoder
- Epigenomic encoder
- Multimodal fusion encoder
- Configurable latent dimension
- Embedding regularization

C. Causal Layer

- Explicit do-operator logic
- Counterfactual generation
- Multi-gene perturbation modeling
- Intervention isolation from learned weights

D. Dynamics Layer

- Graph Neural Network propagation
- Message passing over gene networks
- Multi-step temporal rollout
- Stability enforcement
- Configurable simulation horizon

E. Uncertainty Layer

- Deep ensembles
- Monte Carlo sampling
- Predictive variance tracking
- Calibration correction
- Time-dependent uncertainty propagation

F. Explainability Layer

- Gene-level attribution
- Pathway-level importance scoring
- Temporal root-cause tracing

- Sensitivity analysis
- Feature influence heatmaps

G. Active Learning Layer

- Information gain estimation
- Exploration vs exploitation balancing
- Budget-aware recommendation
- Multi-perturbation ranking

H. Application Layer

- REST API
- Web-based dashboard
- Simulation interface
- Scenario comparison interface
- Results export
- Configurable experiment environment

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6. System Architecture

ATLAS follows a 7-layer modular architecture:

1. Data Layer
2. Representation Layer
3. Causal Layer
4. Dynamics Layer
5. Trust Layer (Uncertainty + Explainability)
6. Decision Layer
7. Application Layer

Each layer is isolated and testable.

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7. Functional Requirements (Complete)

7.1 Data Management

- Support structured omics input
- Preprocessing pipeline with QC
- Config-driven dataset loading

- Dataset version tracking

7.2 Encoding

- Support variable input gene counts
- Modular encoder swapping
- Latent state export

7.3 Intervention Handling

- Accept single and combinatorial gene perturbations
- Modify latent state without retraining
- Maintain intervention trace logs

7.4 Simulation Engine

- Deterministic rollout
- Configurable number of steps
- State logging at each timestep
- Prevent exploding trajectories

7.5 Uncertainty Estimation

- Provide mean and variance outputs
- Time-step-specific confidence intervals
- Export uncertainty metrics

7.6 Explainability

- Rank genes by influence
- Map genes to biological pathways
- Provide temporal attribution heatmaps

7.7 Active Learning

- Rank candidate interventions
- Estimate information gain
- Apply experimental constraints
- Generate top-K recommendations

7.8 API

Endpoints:

- /simulate
- /explain
- /recommend
- /health
- /metadata

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8. Non-Functional Requirements

Category	Requirement
Performance	<5s for moderate graph
Scalability	Dockerized deployment
Reproducibility	YAML config-based runs
Reliability	Offline-trained models only
Observability	Logging and metrics
Maintainability	Layered modular architecture
Extensibility	Plug-and-play modules
Security	Input validation & rate limiting

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9. Data Requirements

Supported Inputs:

- scRNA-seq
- ATAC-seq
- Perturb-seq datasets
- Regulatory interaction networks
- Pathway databases

Data Formats:

- AnnData
- CSV
- JSON
- Graph adjacency matrices

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10. Tech Stack

AI/ML

- Python
- PyTorch
- PyTorch Geometric

Bioinformatics

- Scanpy
- AnnData

Backend

- FastAPI
- Pydantic

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- D3 / Cytoscape for visualization

Infrastructure

- Docker
- Docker Compose
- YAML configs
- GitHub Actions CI

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11. Model Lifecycle

1. Data ingestion
2. Offline training
3. Validation & calibration
4. Artifact freezing
5. Containerized deployment
6. Inference-only serving

No live training.

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12. Evaluation Framework

- Predictive error
- Calibration error
- Stability metrics
- Attribution robustness
- Recommendation quality
- Simulation realism validation

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13. Risk Assessment

Risk	Mitigation
Model overfitting	Regularization + ensemble
Unstable simulation	Rollout constraints
Miscalibration	Temperature scaling
Biological simplification	Graph constraints
Computational cost	Configurable model depth

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14. Success Metrics

- Accurate perturbation predictions
- Stable multi-step rollouts
- Calibrated uncertainty
- Interpretable explanations
- Practical experiment recommendations
- Reproducible containerized system

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15. Deliverables

- Complete source repository
- Trained model artifacts
- Deployment container
- Full documentation
- Research manuscript
- Workflow diagrams
- Architecture diagrams

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16. Long-Term Extensions

- Structural causal discovery
- Multi-cell population simulation
- Reinforcement learning experiment planning
- Cloud-scale distributed training
- Cross-species modeling

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17. System Definition

ATLAS is a scalable, intervention-aware, uncertainty-calibrated causal simulation and decision-support platform for modeling single-cell perturbation dynamics.